DIVYANSHU GOYAL

Multimodal Model Training | RL Fine-Tuning | Evaluation

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SUMMARY

Applied scientist with **8+ years at Adobe**, passionate about multimodal large language models. My work spans two areas: building agentic AI products, and training, aligning, and evaluating multimodal and large language models in subjective domains such as marketing and photo quality. As tech lead on a team of 20+, I worked on taking **Adobe GenStudio for Performance Marketing from prototype to a multi-million-dollar product** used by Fortune 500 companies. Today I am working as a tech lead for a 7-person team building marketing VLMs, judge models, and photo-retouching models for domains where quality depends on human taste. With Adobe Research, I co-led **ATune, a 54M-parameter retoucher that human raters prefer to GPT-6 Astra**. My work has been published at CVPR, ICML, and AAAI workshops and I am an inventor on 6 patents. I am a strong believer in first-principles thinking.

EDUCATION

Institution	Degree	GPA	Years
Georgia Institute of Technology	M.S. Computer Science (Machine Learning)	4.0/4.0	2019 – 2021
BITS Pilani	B.E. Computer Science, M.Sc. Mathematics	8.75/10	2011 – 2016

EXPERIENCE

Applied Scientist – Adobe Inc., San Jose, CA — June 2021 – Present

FixIt: Multimodal Models for Marketing and Photo Quality (Tech Lead, team of 7)

- Tech lead for three multimodal models: a marketing VLM, judge models, and the ATune retoucher. Defined the data annotation strategy, the training pipeline, and the evaluation protocol, keeping training stable and fast and verifying that each model learned its intended reward.
- Demonstrated **sub-second** decisions on marketing content. The demos secured **org-wide investment** in these models and led to customer engagements that **opened new revenue streams**.

Adobe Marketing VLM (Gemma-3-12B, SFT and GRPO)

- Fine-tuned **Gemma-3-12B** with distributed SFT followed by multi-round GRPO on 8× H100, using LoRA at rank 256 (1.19B trainable parameters). Each GRPO round samples a group of answers per prompt, scores them against ground truth, and updates the policy with a clipped policy-gradient loss.
- Designed programmatic rewards for 7 tasks: marketing quality, distortion type and severity, emotion, audience, attention, creative attributes, and photo observables.
- Diagnosed **reward hacking**. Rollout audits showed the policy collapsing onto hedged answers: it gave the middle quality rating to 84% of images, against 56% for human reviewers. Traced the cause to the reward's shape and redesigned the scorers.
- Built the data and serving stack: an asynchronous VLM annotation pipeline with chunked checkpoints and crash recovery, and a router-and-worker inference API across 4 GPUs.

JEV-Style Marketing Judge Models

- Trained Qwen3.5-4B LoRA judges on a **1.1M-example dataset** curated from 14 human-rated sources, using the spread of rater opinions as soft labels.
- Raised correlation with human ratings on KonIQ to **0.663 SRCC**, up from 0.421 for the reference model, after calibrating the score output on crowd ratings (paired 95% CI above zero).
- Built a **67K-item held-out benchmark**, leak-checked with SSCD copy detection against all 537K training images. It gated every release from v9 to v19 through a pre-registered non-inferiority test. Served the judge at 21 replicas across 8 H100s.
- Applied the judge to pairwise crop selection, reaching **0.82 accuracy** on GAIC vs. 0.71 for the score-based baseline.
- Detected a run whose vision encoder collapsed within 200 steps, leaving every checkpoint blind to the image, and added a vision-health check to the training pipeline.

ATune: Unsupervised Photo Retouching (Co-Lead with Adobe Research)

- Co-led the project with Adobe Research as co-first author (under review). Mentored an intern and owned the synthetic data pipeline, the training recipe, and the evaluation.
- Trained a **54M-parameter model** that retouches photos by setting the sliders of off-the-shelf editors (RawTherapee and darktable). It learns from **10M unedited photographs, with no expert edits**.
- In a human study with 157 raters and 3,524 judgments, its edits **beat GPT-6 Astra** using the same editor (54.3% of decisive votes, 95% CI 50.3–58.3) and **matched Claude Opus 5** (50.5%). On FiveK, PPR10K, and SPAQ, at least 8 of 9 published image-quality judges also prefer it to GPT-6 Astra.
- Ran a data-scaling study from 1M to 10M photographs: each larger corpus wins or ties, and no judge on any test set prefers a smaller one.

Adobe GenStudio for Performance Marketing (Tech Lead, team of 20+)

- Took GenStudio from **prototype to a multi-million-dollar product** used by **Fortune 500 companies** (launched October 2024). Tech lead for the production API server that connects the product to cloud LLM providers at scale.
- Co-invented a system that uses generative models to turn a company's source material into structured brand guidelines, generate marketing content from them, and score that content against each guideline (US20250117829). Designed the brand-compliance evaluation; the work became the product **Adobe Brand Intelligence**.
- Fine-tuned custom models on a Fortune 500 customer's own data, which led to a **custom-models deal**.
- Built a multimodal agent that edits images in Photoshop, selecting from 49 tools with vision grounding.

Evaluation Research

- **DistortBench** (first author; CVPR and ICML 2026 workshops): a benchmark of 13.5K questions over 27 distortion types and 5 severity levels. The best of 18 VLMs (Qwen3.5-27B, 61.9%) beats the average annotator (60.2%) but trails a 3-annotator majority vote (65.7%).
- **JudgeBreak** (ICML 2026 workshop): showed that linear probes on Qwen3-32B hidden states beat the best reward model or LLM judge across 95 subjective settings (78.1% vs. 70.1% pairwise accuracy).

Machine Learning Engineer – Adobe Inc., Bangalore, India — July 2016 – August 2019

- Joined Adobe's ML serving platform at inception and helped plan the end-to-end product, which hosts internal models such as field-aware factorization machines.
- Owned serving throughput: scaled the production ML prediction service to **3,500 QPS per node at 0.3–0.6 ms p99 latency** (Adobe Tech Blog), with a garbage-collection-free asynchronous event pipeline on LMAX Disruptor that removed back-pressure under peak load.
- Built the model lifecycle platform for versioned deployment, monitoring, and staged rollout across environments.

PUBLICATIONS

ATune: Unsupervised Photo Retouching Through Image-Editor Controls
D. Goyal*, G. Ruiz*, A. Meisami | Under review | *equal contribution

DistortBench: Benchmarking Vision Language Models on Image Distortion Identification
D. Goyal, A. Eppa, V. Bannihatti Kumar | CVPR 2026 and ICML 2026 Workshops | arXiv:2604.19966

JudgeBreak: Is Taste Hidden in the Layers? Stress-Testing Reward Models and LLM Judges with Representation-Based Subjective Evaluation
V. Bannihatti Kumar, A. Narang, M. Anand, A. Eppa, **D. Goyal** | ICML 2026 Workshop | OpenReview

Curiosity-Driven LLM-as-a-judge for Personalized Creative Judgment
V. Bannihatti Kumar, **D. Goyal**, A. Eppa, N. Bhandari | AAAI 2026 Workshop, Oral | arXiv:2510.05135

PATENTS

Title	Number	Year
Guiding Language Translation with Translation Documents Using ML	US20260050753	2026
Brand-Aligned Marketing Content Generation (Adobe Brand Intelligence)	US20250117829	2025
Cultural Adaptation of Images Using Generative AI	US20260051084	2026
Parameter-Efficient Fine-Tuned Models	US20250217193	2025
Video-to-Video Culturalization Using Generative AI	Approved for filing	2026
Learning to Be Surprised: A Curiosity-Driven Framework for Creative Judgment	Approved for filing	2026

EXTERNAL CONTRIBUTIONS

VibeNanoChat: GPT-2 Scale LLM Pretraining from Scratch — 2026 | code
Pretrained GPT-2-scale transformers from scratch at **over 1M tokens/s** on 8× H100, with a custom distributed Muon optimizer. Derived compute-optimal model sizes from isoFLOP curves over 40+ runs. Write-ups: Going Fast and A Lazy Engineer's Guide to Scaling Transformers.

BrownBearAI: On-Device Photo Understanding for iOS — 2026 | App Store
Shipped an iPhone app for on-device face clustering and duplicate detection, reaching **99.4% BCubed F1** on LFW identities.

LaSeRo: Segmenting Antarctic Rock Outcrops from Landsat 8 — 2019–2020 | code
Trained and evaluated SegNet models to segment rock outcrops in multispectral satellite imagery.

Face Mask Detection for COVID-19 — 2020 | code
Built a mask classifier during the pandemic, comparing four CNN architectures with VAE pretraining and Grad-CAM attribution.

TECHNICAL SKILLS

Research: VLM Training & Evaluation, LLM Pretraining & Scaling Laws, Benchmark Design, Human Preference Studies, LLM-as-a-Judge & Reward Models, Reinforcement Learning (GRPO), Reward Design, Linear Probing, Bootstrap Statistics

Tools: Python, PyTorch, Distributed Training (DDP, ZeRO-2), torch.compile, FlashAttention-3, Liger Kernels, Muon, Mixed Precision (BF16), LoRA, Hugging Face Transformers/PEFT, Weights & Biases